

Question_2:

Code:

```
1  /*
2  Question No: 2
3  Remove the smallest and largest elements and insert their
4  difference. After K operations, print the sum of the array.
5  Each query starts from the original array.
6  */
7
8  #include <iostream>
9  #include <vector>
10 #include <set>
11 using namespace std;
12
13 int main(){
14     int n,q;
15     cin>>n>>q;
16     multiset<long long> s;
17     long long sum=0;
18     for(int i=0;i<n;i++){
19         long long x;
20         cin>>x;
21         s.insert(x);
22         sum+=x;
23     }
24     vector<long long> ans(n);
25     ans[0]=sum;
26     for(int i=1;i<n;i++){
27         auto mn=s.begin();
28         auto mx=prev(s.end());
29         long long a=*mn,b=*mx;
30         s.erase(mn);
31         s.erase(mx);
32         s.insert(b-a);
33         sum-=a+b;
34         sum+=b-a;
35         ans[i]=sum;
36     }
37     while(q--){
38         int k;
39         cin>>k;
40         cout<<ans[k]<<"\n";
41     }
42 }
```

Output:

Test Case_1:

```
5 2
3 2 1 5 4
1
13
2
9
PS C:\Users\Prabin Yadav\
```

Test Case_2:

```
5 3
13 12 11 15 14
1
43
2
35
3
15
PS C:\Users\Prabin Yadav\
```